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Evan Roman

Relevant Experience

May 2022 – Present
Software and Systems Engineer, *Loikka*, Turku, Finland

Automated Testing Engineer (Maritime Sector) *Aug 2024 – Present*

- Designed, implemented, and maintained comprehensive automated acceptance test suites using Cucumber and Gherkin, executed on an air-gapped, on-premise maritime system to support formal Customer Acceptance Testing.
- Built an automated provisioning system using Ansible to configure physical offline hardware, ensuring reliable and repeatable test execution.
- Engineered and maintained a virtualized replica of the system running on Kubernetes within Azure DevOps CI pipelines, maintaining test parity between software versions and operational environments.
- Enabled a proprietary DDSI-RTPS (v2.5) dialect to operate reliably within Kubernetes container networks; used Wireshark for packet-level protocol analysis to diagnose Simple Endpoint Discovery Protocol (SEDP) anomalies and deviations from the OMG standard.

SaaS & Cloud Applications *May 2022 – Aug 2024*

- Developed a SaaS platform for subscription and insurance policy management, alongside client applications running on Google Cloud and Google Kubernetes Engine (GKE).
- Implemented CI pipelines in self-hosted GitLab for QA, testing, and deployment to Google Kubernetes Engine with Helm and Pulumi.
- Created and maintained ETL pipelines that run as Kubernetes Jobs with ArangoDB and BigQuery.
- Developed collaborative real-time web applications using WebSockets, Yjs, and Redis Pub/Sub.
- Developed front-end applications in TypeScript with React and Next.js, and back-end APIs with Node.js (Fastify, Express).

Oct 2023 – Dec 2023
Teaching Assistant, *Åbo Akademi University*, Turku, Finland

I gave lectures for a master's course in Computer Engineering.

May 2021 – Sep 2021
Developer, *Digital Onboarding*, Boston, MA

I worked on a team improving a marketing tool for banks and credit unions.

- Developed back-end API in Elixir using Phoenix and Ecto with PostgreSQL.
- Developed front-end SPA in React and Redux.

Dec 2018 – Jul 2019
Technical Co-founder, *SAM*, New York, NY

Co-founded early-stage startup, architecting the initial MVP, API specification, and cloud infrastructure.

- Designed API with founders in OpenAPI on Swagger.
- Developed front-end for SPA in React and back-end in Django.
- Created DevOps system with Docker and provisioned on AWS.

Jan 2017 – Jan 2018
University Lecturer, *Tallinn University of Technology*, Tallinn, Estonia

I taught "Fundamentals of Python" and "Advanced Python" to bachelor's students.

- Tested students' ability to use primitive data types, lists, and dictionaries using unit tests.
- Gave lectures and exercises on Python features including concurrency with threads and asyncio, decorators, iterators, generators, and comprehensions.
- Created challenges and aided students with popular libraries, like Django, Flask, NumPy, Pandas, Requests, and Scikit-learn.

Oct 2015 – Feb 2017 **Lead Developer**, *Like A Local Guide*, Tallinn, Estonia

I took full responsibility for the maintenance and development of "likealocalguide.com". I furthered the company's vision of the project, evolving the website to a more social platform. As the main steward of the project, I led other developers who also worked on the project.

- Implemented sales funnels from Google Analytics, identifying steps where users dropped off when booking tours or adding content.
- Extracted, Transformed, and Loaded (ETL) over 100,000 cities from GeoNames, while deduplicating and matching GeoNames records with existing database entries.
- Migrated deployment from Arch Linux to CentOS.
- Increased site speed on pages by caching expensive operation results with Redis and query optimization with PostgreSQL.
- Maintained CentOS server and CentOS Vagrant development environment.
- Scaled product by adding PgBouncer, upgrading server hardware, and appropriately tuning the number of worker processes for the new hardware.
- Implemented designs and features on the front-end with SASS and jQuery and on the back-end with Django.
- Interviewed and managed other developers working on the project.

May 2015 **Developer**, *Cakebet*, San Francisco, CA

I contracted as a developer for a Bitcoin casino.

- Built a currency converter covering most fiat and numerous cryptocurrencies, updating at high frequency (<3 seconds).

Education

Sep 2021 – Jun 2023 **MS Computer Science**, *Åbo Akademi University*, Turku, Finland

- Received full tuition merit scholarship.
- Thesis: *Performance of Special-Interest Voting Advice Applications in the 2023 Finnish Parliamentary Election*.
  https://www.doria.fi/handle/10024/187299

May 2018 – Mar 2020 **BA Mathematics**, *Thomas Edison State University*, Trenton, NJ

Bachelor's project: *Explaining Black Boxes: Interpretable Machine Learning*. GPA 3.8/4.0

Skills

Programming Languages: C++, Python, TypeScript, JavaScript (ES6+), Elixir, Rust, Shell (POSIX, bash), Gherkin, Emacs Lisp, Matlab/Octave, R.

Frameworks & Libraries: Cucumber, React, Next.js, Yjs, Phoenix, Ecto, Fastify, Express.js, Django, Flask, Pandas, NumPy, D3.js.

Systems & Protocols: DDS (DDSI-RTPS v2.5, SEDP), Wireshark, OpenAPI (Swagger).

DevOps & Infrastructure: Kubernetes (kind, GKE), Docker, Docker Compose, Ansible, Azure DevOps, GitLab CI/CD, Helm, Pulumi, Google Cloud, AWS, Linux (RHEL, Debian).

Databases: PostgreSQL, MySQL, SQLite, Redis, BigQuery, MongoDB, ArangoDB.

Projects and Open Source

Grid Validator, *https://evrom.github.io/grid/*

Validates, previews, and helps debug `grid-template-areas` CSS property values.

- Built Grid Validator with TypeScript and React.
- https://github.com/evrom/grid-validator

Stylelint, *https://github.com/stylelint/stylelint/*

Added a new validation rule to Stylelint.

- Validates `named-grid-areas-no-invalid` against W3C specifications.
- Provides descriptive error messages identifying non-contiguous or non-rectangular values within template strings.
- https://github.com/stylelint/stylelint/pull/5167

Hurst, *https://crates.io/crates/hurst*

Calculates the estimated Hurst exponent of time series data.

- Published WebAssembly package on NPM.
- Written in Rust.
- https://github.com/evrom/hurst

Hurst Exponent, *https://evrom.github.io/hurst-exponent/*

Interactive visualization of the estimated Hurst exponent of time series data.

- Built with TypeScript, React, and D3.js.
- Uses Hurst compiled to WebAssembly.
- https://github.com/evrom/hurst

This Resume, *https://github.com/evrom/resume*

View the source of this resume in LaTeX.